

SANGMIN WOO

Applied Scientist, AWS Agentic AI

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EXPERIENCE

Amazon Web Services (AWS) Applied Scientist, Agentic AI Applied Scientist Intern, Bedrock (<i>remote from Sep 2024</i>)	<i>Santa Clara, CA, USA</i> Jun 2025 – Present Jun 2024 – Mar 2025
NAVER LABS Research Intern, Robot Vision Team	<i>Suwon, Korea</i> Apr 2023 – Aug 2023

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Ph.D. in Electrical Engineering · Advisor: Changick Kim	<i>Daejeon, Korea</i> Aug 2021 – Aug 2025
Gwangju Institute of Science and Technology (GIST) M.S. in Electrical Engineering and Computer Science · Advisor: Kangil Kim	<i>Gwangju, Korea</i> Sep 2019 – Aug 2021
Kyungpook National University (KNU) B.S. in Electrical Engineering	<i>Daegu, Korea</i> Mar 2013 – Aug 2019

PROJECTS

Skill Evaluators · Amazon Bedrock AgentCore Evaluations & Strands Agents Evaluators for skill-equipped agents that measure whether an agent selects the appropriate skills and faithfully follows their prescribed instructions, with trajectory-based evaluation across multiple agent frameworks.	2026 AgentCore Strands Blog
Multimodal Evaluators · Strands Agents MLLM-as-a-judge evaluators for image-to-text tasks that score overall quality, correctness, faithfulness, and instruction following, in both reference-based and reference-free settings.	2026 Strands Blog
Agent Evaluators · Amazon Bedrock AgentCore Evaluations Automated service for continuously assessing AI agent quality, measuring end-to-end task success, tool-use accuracy, and other built-in or custom metrics over live interactions for real-time monitoring.	2025 AgentCore
Evaluation Agent · AWS Labs & Strands Agents AI-driven toolkit that automates end-to-end agent evaluation: planning, test-case generation, trace collection, metric implementation, execution, analysis, and actionable improvement recommendations.	2025 AWS Labs Strands Blog

PUBLICATIONS

<i>International Conferences, Journals & Preprints</i>	* Equal contribution
[30] StructRL: Online Structured Reinforcement Learning for Long-Horizon Vision-Language-Action Tasks Ziyi Yin*, Sangmin Woo*, Kang Zhou, Sungyeon Kim, Aosong Feng, Haibo Ding, Luke Huan	<i>arXiv 2026</i> Paper Code Project
[29] An Empirical Study of Automating Agent Evaluation Kang Zhou*, Sangmin Woo*, Haibo Ding, Kiran Ramnath, Aosong Feng, Subramanian Chidambaram, Vivek Singh, Vinayak Arannil, Muhyun Kim, Ishan Singh, Darren Wang, Zhichao Xu, Megha Gandhi, Nirmal Prabhu, Soumya Smruti Mishra, Gouri Pandeshwar, Lin Lee Cheong	<i>arXiv 2026</i> Paper Code Blog
[28] Beyond Correctness: Rewarding Faithful Reasoning in Retrieval-Augmented Generation Zhichao Xu*, Zongyu Wu*, Yun Zhou, Aosong Feng, Kang Zhou, Sangmin Woo, Kiran Ramnath, Yijun Tian, Xuan Qi, Weikang Qiu, Lin Lee Cheong, Haibo Ding	<i>TMLR 2026 · KnowFM @ ACL 2026</i> Paper
[27] Learning to Ideate for Machine Learning Engineering Agents Yunxiang Zhang, Kang Zhou, Zhichao Xu, Kiran Ramnath, Yun Zhou, Sangmin Woo, Haibo Ding, Lin Lee Cheong	<i>EACL 2026 · Oral</i> Paper
[26] Diffusion Language Model Inference with Monte Carlo Tree Search Zheng Huang, Kiran Ramnath, Yueyan Chen, Aosong Feng, Sangmin Woo, Balasubramaniam Srinivasan, Zhichao Xu, Kang Zhou, Shuai Wang, Haibo Ding, Lin Lee Cheong	<i>Findings of EACL 2026</i> Paper
[25] What and When to Look?: Temporal Span Proposal Network for Video Relation Detection Sangmin Woo, Junhyug Noh, Kangil Kim	<i>ESWA 2025</i> Paper Code

- [24] **A Systematic Survey of Automatic Prompt Optimization Techniques** *EMNLP 2025*
Kiran Ramnath, Kang Zhou, Sheng Guan, Soumya Smruti Mishra, Xuan Qi, Zhengyuan Shen, Shuai Wang, Sangmin Woo, Sullam Jeoung, Yawei Wang, Haozhu Wang, Han Ding, Yuzhe Lu, Zhichao Xu, Yun Zhou, Balasubramaniam Srinivasan, Qiaojing Yan, Yueyan Chen, Haibo Ding, Panpan Xu, Lin Lee Cheong *Paper*
- [23] **Don't Miss the Forest for the Trees: Attentional Vision Calibration for Large Vision Language Models** *Findings of ACL 2025*
Sangmin Woo*, Donguk Kim*, Jaehyuk Jang*, Yubin Choi, Changick Kim *Paper | Code | Project*
- [22] **Modality Mixer Exploiting Complementary Information for Multi-modal Action Recognition** *CVIU 2025*
Sumin Lee, Sangmin Woo, Yeonju Park, Muhammad Adi Nugroho, Changick Kim *Paper*
- [21] **Parameter Efficient Mamba Tuning via Projector-targeted Diagonal-centric Linear Transformation** *CVPR 2025*
Seokil Ham, Hee-Seon Kim, Sangmin Woo, Changick Kim *Paper*
- [20] **Black-Box Visual Prompt Engineering for Mitigating Object Hallucination in Large Vision Language Models** *NAACL 2025*
Sangmin Woo, Kang Zhou, Yun Zhou, Shuai Wang, Sheng Guan, Haibo Ding, Lin Lee Cheong *Paper*
- [19] **Diffusion Model Patching via Mixture-of-Prompts** *AAAI 2025*
Seokil Ham*, Sangmin Woo*, Jinyoung Kim, Hyojun Go, Byeongjun Park, Changick Kim *Paper | Code | Project*
- [18] **RITUAL: Random Image Transformations as a Universal Anti-hallucination Lever in LVLMS** *arXiv 2024*
Sangmin Woo*, Jaehyuk Jang*, Donguk Kim*, Yubin Choi, Changick Kim *Paper | Code | Project*
- [17] **Flow-Assisted Motion Learning Network for Weakly-Supervised Group Activity Recognition** *ECCV 2024*
Muhammad Adi Nugroho, Sangmin Woo, Sumin Lee, Jinyoung Park, Yooseung Wang, Donguk Kim, Changick Kim *Paper*
- [16] **Spatio-Temporal Proximity-Aware Dual-Path Model for Panoramic Activity Recognition** *ECCV 2024*
Sumin Lee, Yooseung Wang, Sangmin Woo, Changick Kim *Paper*
- [15] **Switch Diffusion Transformer: Synergizing Denoising Tasks with Sparse Mixture-of-Experts** *ECCV 2024*
Byeongjun Park, Hyojun Go, Jinyoung Kim, Sangmin Woo, Seokil Ham, Changick Kim *Paper | Code | Project*
- [14] **HarmonyView: Harmonizing Consistency and Diversity in One-Image-to-3D** *CVPR 2024*
Sangmin Woo*, Byeongjun Park*, Hyojun Go, Jinyoung Kim, Changick Kim *Paper | Code | Project | Demo*
- [13] **Denoising Task Routing for Diffusion Models** *ICLR 2024*
Byeongjun Park*, Sangmin Woo*, Hyojun Go*, Jinyoung Kim*, Changick Kim *Paper | Code | Project*
- [12] **Sketch-based Video Object Localization** *WACV 2024*
Sangmin Woo, Soyeong Jeon, Jinyoung Park, Minji Son, Sumin Lee, Changick Kim *Paper | Code*
- [11] **AHFu-Net: Align, Hallucinate, and Fuse Network for Missing Multimodal Action Recognition** *VCIP 2023 · Oral*
Muhammad Adi Nugroho, Sangmin Woo, Sumin Lee, Changick Kim *Paper*
- [10] **Multi-modal Social Group Activity Recognition in Panoramic Scene** *VCIP 2023*
Donguk Kim, Sumin Lee, Sangmin Woo, Jinyoung Park, Muhammad Adi Nugroho, Changick Kim *Paper*
- [9] **Cross-Modal Alignment and Translation for Missing Modality Action Recognition** *CVIU 2023*
Yeonju Park, Sangmin Woo, Sumin Lee, Muhammad Adi Nugroho, Changick Kim *Paper*
- [8] **Audio-Visual Glance Network for Efficient Video Recognition** *ICCV 2023*
Muhammad Adi Nugroho, Sangmin Woo, Sumin Lee, Changick Kim *Paper*
- [7] **Towards Good Practices for Missing Modality Robust Action Recognition** *AAAI 2023 · Oral*
Sangmin Woo, Sumin Lee, Yeonju Park, Muhammad Adi Nugroho, Changick Kim *Paper | Code*
- [6] **Modality Mixer for Multi-modal Action Recognition** *WACV 2023*
Sumin Lee, Sangmin Woo, Yeonju Park, Muhammad Adi Nugroho, Changick Kim *Paper*
- [5] **Explore-And-Match: Bridging Proposal-Based and Proposal-Free with Transformer for Sentence Grounding in Videos** *arXiv 2022*
Sangmin Woo, Jinyoung Park, Inyong Koo, Sumin Lee, Minki Jeong, Changick Kim *Paper | Code*
- [4] **Tackling the Challenges in Scene Graph Generation with Local-to-Global Interactions** *IEEE TNNLS 2022*
Sangmin Woo, Junhyug Noh, Kangil Kim *Paper | Code*
- [3] **Temporal Flow Mask Attention for Open-Set Long-Tailed Recognition of Wild Animals in Camera-Trap Images** *ICIP 2022*
Jeongsoo Kim, Sangmin Woo, Byeongjun Park, Changick Kim *Paper*
- [2] **Impact of Sentence Representation Matching in Neural Machine Translation** *Applied Sciences 2022*
Heeseung Jung, Kangil Kim, Jong-Hun Shin, Seung-Hoon Na, SangKeun Jung, Sangmin Woo *Paper*
- [1] **Revisiting Dropout: Escaping Pressure for Training Neural Networks with Multiple Costs** *Electronics 2021*
Sangmin Woo, Kangil Kim, Junhyug Noh, Jong-Hun Shin, Seung-Hoon Na *Paper | Code*

Korean Domestic Conferences

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|------|---|------------------|
| [D5] | Mitigating Hallucinations in Vision Language Models with High-pass filter
Donguk Kim, Sangmin Woo, Jaehyuk Jang, Yubin Choi, Changick Kim | IEIE 2024 |
| [D4] | Contrastive Decoding with Image Processing Filters for Mitigating Object Hallucinations in Large-scale Vision-Language Models
Yubin Choi, Donguk Kim, Sangmin Woo, Jaehyuk Jang | IEIE 2024 |
| [D3] | Light-Weighted Korean Speech Recognition System for Edge Devices
Yooseung Wang, Sangmin Woo, Changick Kim | IEIE 2023 |
| [D2] | On Learning Relations between Objects in Images
Sangmin Woo, Changick Kim | KIMST 2022 |
| [D1] | Effective Trash Classification using Attentional Learning
Sangmin Woo, Soon Ki Jung | KSC 2018
Code |

PATENTS

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|-----|---|---------------|
| [5] | Model Adaptive Visual Prompt Routing System
Sangmin Woo, Kang Zhou, Yun Zhou, Shuai Wang, Sheng Guan, Haibo Ding, Lin Lee Cheong | US Patent |
| [4] | Group Activity Recognition Apparatus and Method using RGB Videos and LiDAR Data
Changick Kim, Jinyoung Park, Donguk Kim, Sumin Lee, Muhammad Adi Nugroho, Sangmin Woo, Yooseung Wang | KR 10-2741168 |
| [3] | Method and Device for Inferring Dynamic Relationship between Objects in Video
Sangmin Woo, Kangil Kim | KR 10-2590916 |
| [2] | Human Activity Recognition Apparatus, Human Activity Recognition Method and Computer Program for Classifying Activity Categories of User
Changick Kim, Inyong Koo, Yeonju Park, Minki Jeong, Sumin Lee, Sangmin Woo | KR 10-2515065 |
| [1] | Scene Graph Generation Apparatus
Sangmin Woo, Kangil Kim | KR 10-2254768 |

HONORS & AWARDS

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|---|----------|
| Outstanding Reviewer , CVPR 2025 (711 of 12,593 reviewers) | Jun 2025 |
| Finalist , Qualcomm Innovation Fellowship Korea 2024 | Dec 2024 |
| Invited Paper Talk , Center for Applied Research in Artificial Intelligence (CARAI) Workshop | Oct 2023 |
| Finalist , 29th HumanTech Paper Award, Samsung Electronics | Dec 2022 |
| Top Award (\$10,000) , LG Electronics Robot Contest, LG Electronics | Dec 2021 |
| Excellence Award (\$500) , Creative Space G A.I. & IoT Makerthon, GIST | Nov 2019 |

ACADEMIC SERVICE

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| Area Chair | ACL Rolling Review (ARR), since Oct 2025 |
| Conference Reviewer | CVPR (2024–), ICCV (2025–), ECCV (2024–), NeurIPS (2024–), ICLR (2024–), ICML (2025–), AAAI (2023–), ACL ARR (2026–) |
| Journal Reviewer | IEEE TPAMI (2025–), IEEE TIP (2025–), IEEE TNNLS (2023–), TMLR (2025–) |